

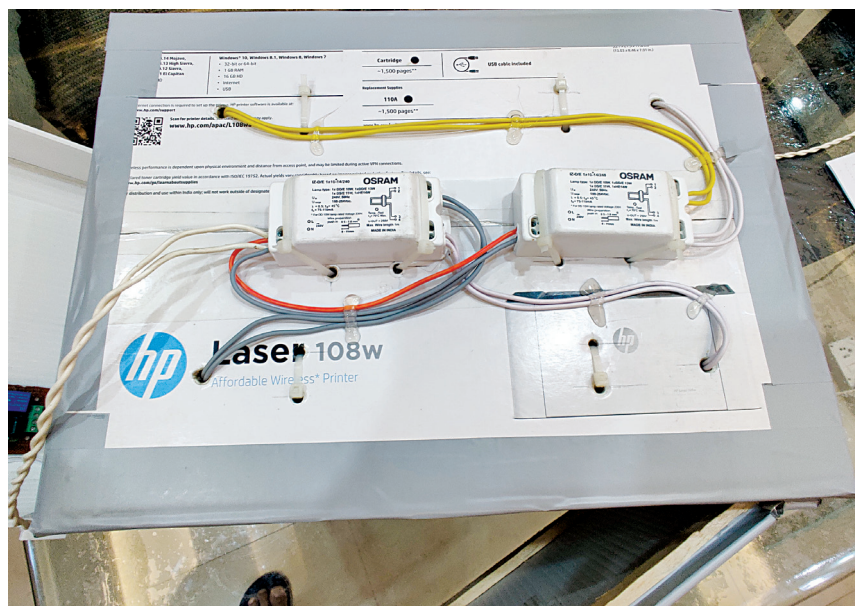


Fig. 3: Installation of the two chokes

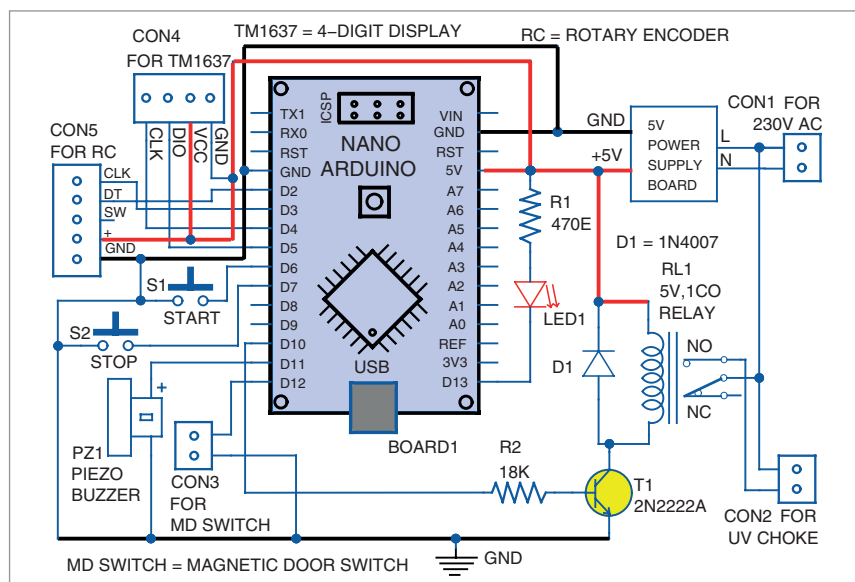


Fig. 4: Circuit diagram

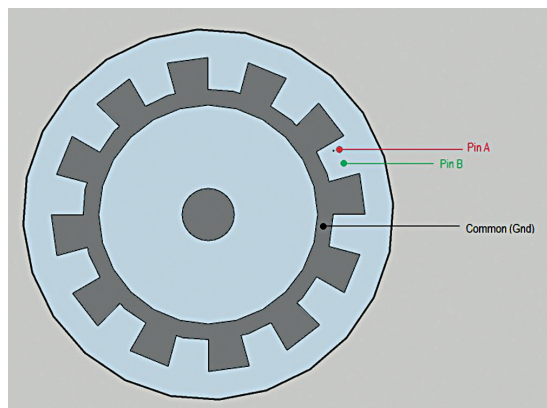


Fig. 5: Illustration of encoder

the side edge of the box to which the magnets can stick.

Stick kitchen aluminium foil or silver craft paper on all inner sides of the box so that the UV light scatters all over inside the box. The outer side of the box can also be covered with craft paper. Install the two 11W UV tubes on ceiling of the box using ties. Draw the end wires connected to the tubes outside the top of the

box where the two chokes are installed as shown in Fig. 3.

Circuit and working

The controller is built around an Arduino. Arduino Pro can be used, which has a small footprint, though any Arduino variant would work. Circuit diagram of the UV sterilisation box is shown in Fig. 4. It comprises a TM1637 module, 5V DC power supply module (Module1), Arduino Nano (Board1), 5V single-changeover relay (RL1), a rotary switch, a magnetic door switch, two 230V AC operated UV tubes, two chokes, and a few other components.

The software sketch can be downloaded from EFY website and uploaded to the Arduino using the Arduino IDE.

A rotary encoder is used for setting time-period. Though it looks like an analogue potentiometer, it is a digital device that can be fully rotated without any end stops. It does not give any absolute value but can be interpreted to give the direction of motion, and the pulses generated can be counted to give the new value. The most common rotary encoder is the scroll wheel on a computer mouse. Just like a mouse, the rotary encoder can be rotated and clicked (pressed). In this circuit, the inbuilt press switch is not used. Inside the encoder there is a slotted disk connected to the common ground pin C, and two contact pins A and B as shown in Fig. 5.

In addition to these three pins, there are two pins for the momentary switch (not used). When the switch is rotated clockwise, pin A gets grounded before pin B, and as the knob is rotated further pin A goes low before pin B. After each full cycle of the pulse, a click can be felt as there are detents (stops). There are 20 such detents in one full rotation.

Similarly, when the knob is rotated counter-clockwise, the pulse sequence changes as pin B goes high first followed by pin A. The pulse train sequence is shown in Fig. 6.

The encoder used here has five pins as shown in Fig. 7. Pin DT of encoder is connected to pin 2 of Arduino while CLK pin is connected to pin D3 of Arduino. The + and GND pins of encoder are connected to 5V and GND pins of Arduino, respectively.